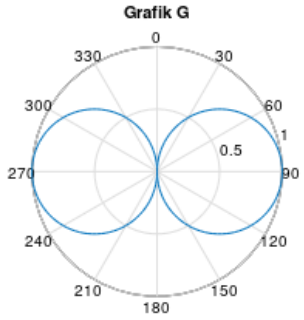
Tugas Besar Antena dan Propagasi

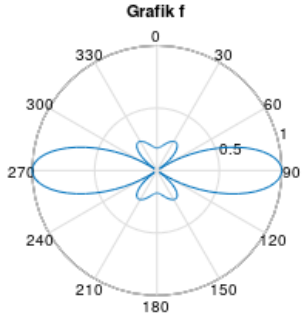
Nama : Fauzan Dwipriyandi (201344012)
Renanda Egy Putra W (201344029)
Kelas : 3NK

1. Sebuah antena susun (array) broadside terdiri dari 4 buah elemen antena dipol ideal. Jika jarak antar elemen $d = \lambda/2$, tentukan dan plot pola elemen pola elemen $g(\theta,\phi)$, pola faktorsusun $f(\theta,\phi)$ dan pola total $F(\theta,\phi)$ untuk masing-masing beda fasa antar elemen α:

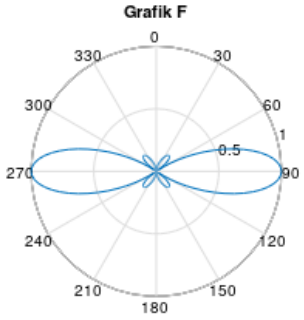
   a) α = 0 °
   - Pola elemen $g(\theta,\phi)$



   - Pola factor susun $f(\theta,\phi)$

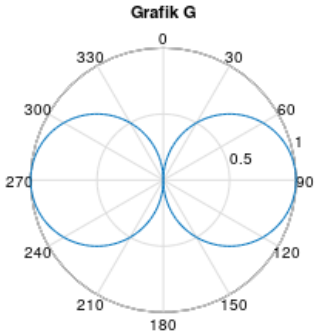


   - Pola total $F(\theta,\phi)$

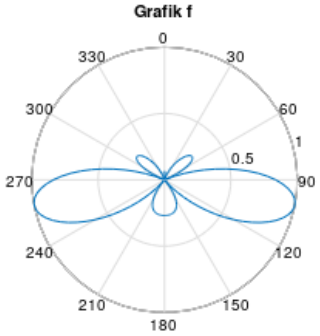


b) α = 30 °

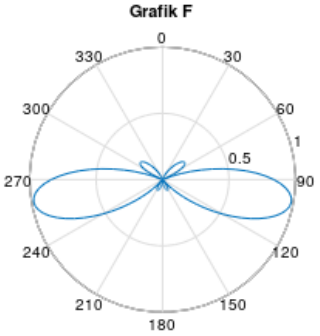
- Pola elemen $g(\theta,\phi)$



- Pola factor susun $f(\theta,\phi)$

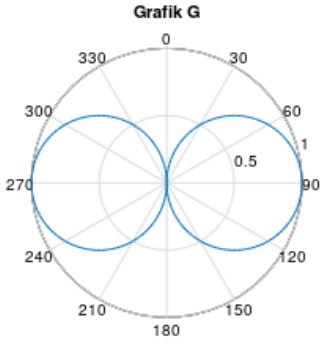


- Pola total $F(\theta,\phi)$

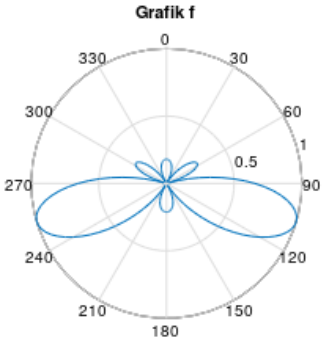


c) α = 45 °

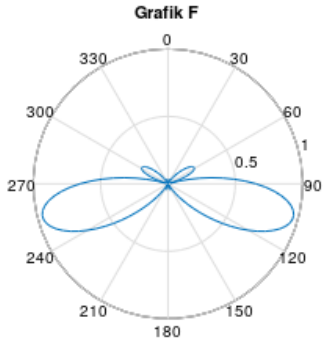
- Pola elemen $g(\theta,\phi)$



- Pola factor susun $f(\theta,\phi)$

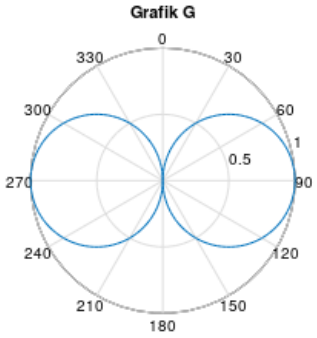


- Pola total $F(\theta,\phi)$

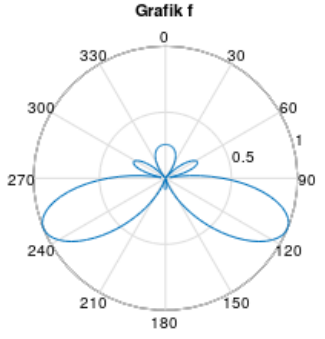


d) α = 60 °

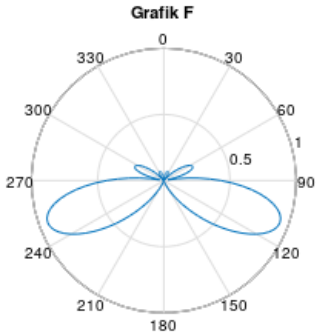
- Pola elemen $g(\theta,\phi)$



- Pola factor susun $f(\theta,\phi)$

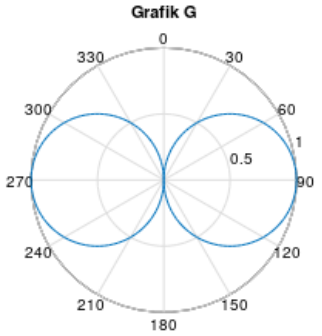


- Pola total $F(\theta,\phi)$

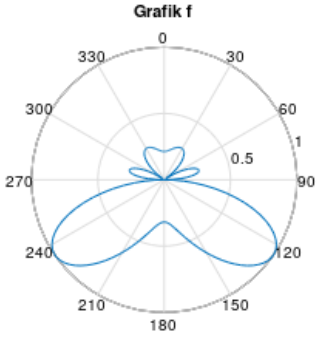


e) α = 90 °

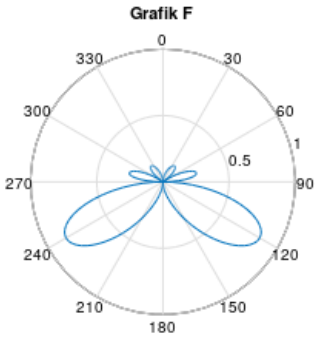
- Pola elemen $g(\theta,\phi)$



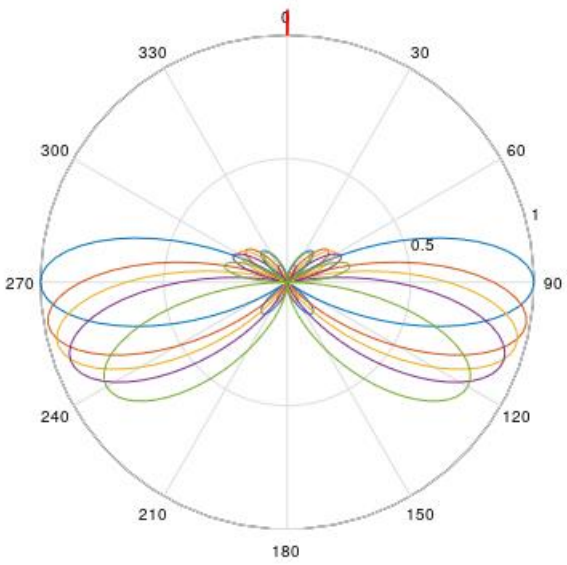
- Pola factor susun $f(\theta,\phi)$



- Pola total $F(\theta,\phi)$

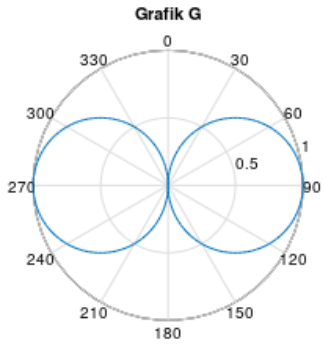


Gambar apabila di plot dalam satu grafik

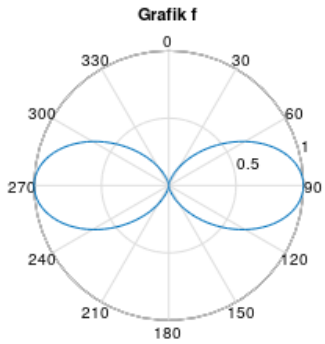


- (Blue) α = 0 °
- (Red) α = 30 °
- (Yellow) α = 45 °
- (Purple) α = 60 °
- (Green) α = 90 °

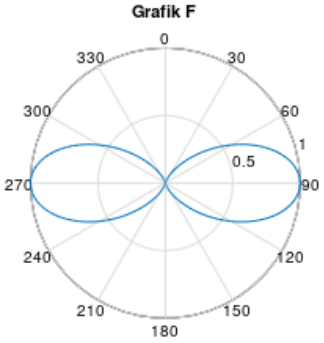
2. Sebuah antena susun (array) broadside terdiri dari 4 buah elemen antena dipol ideal. Jika beda fasa antar elemen α = 0°, tentukan dan plot pola elemen pola elemen $g(\theta,\phi)$, pola faktor susun $f(\theta,\phi)$ dan pola total $F(\theta,\phi)$ masing-masing untuk jarak antar elemen d:
   a) d = $\lambda/4$
      - Pola elemen $g(\theta,\phi)$



      - Pola factor susun $f(\theta,\phi)$

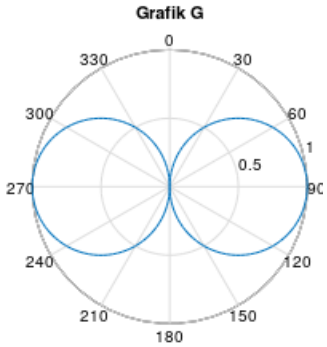


- Pola total $F(\theta,\phi)$

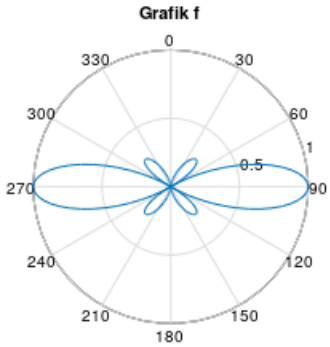


b) d = $\lambda/2$

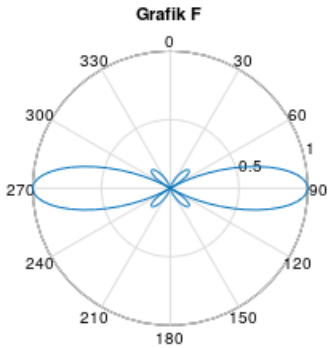
- Pola elemen $g(\theta,\phi)$



- Pola factor susun $f(\theta,\phi)$

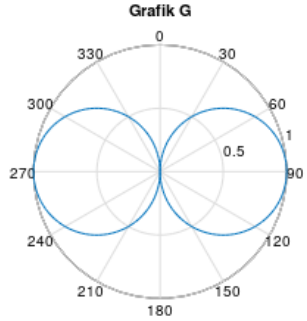


- Pola total $F(\theta,\phi)$

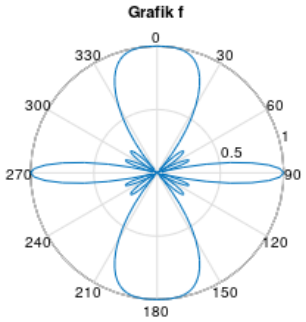


c) d = λ

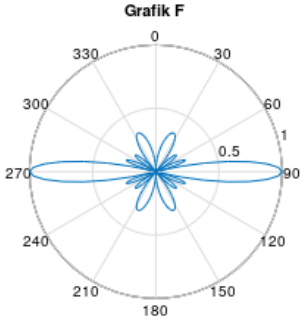
- Pola elemen $g(\theta,\phi)$



- Pola factor susun $f(\theta,\phi)$

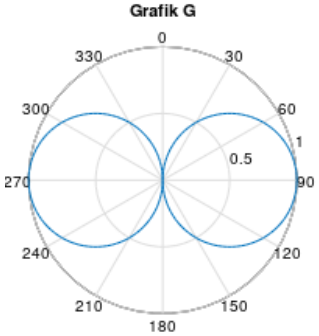


- Pola total $F(\theta,\phi)$

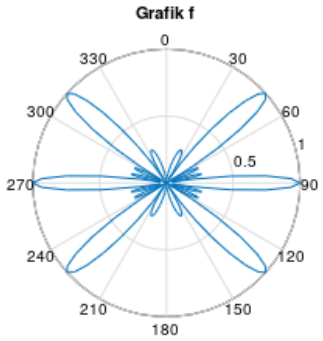


d) d = 3λ/2

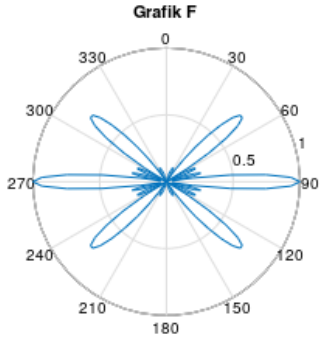
- Pola elemen $g(\theta,\phi)$



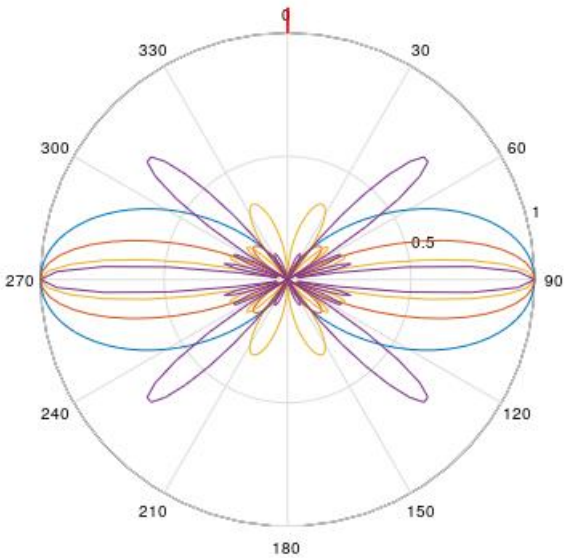
- Pola factor susun $f(\theta,\phi)$



- Pola total $F(\theta,\phi)$

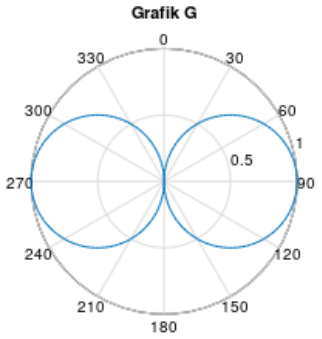


Gambar apabila di plot dalam satu grafik

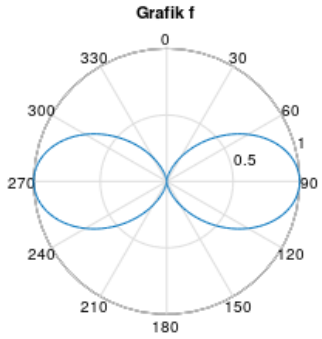


- (Blue) d = $\lambda/4$
- (Red) d= $\lambda/2$
- (Yellow) $\alpha = \lambda$
- (Purple) $\alpha = 3\lambda/2$

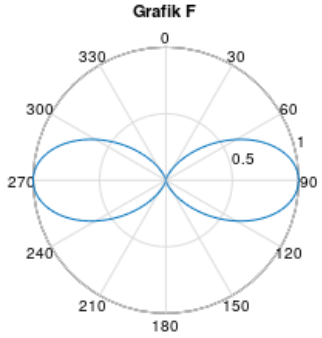
3. Sebuah antena susun (array) broadside terdiri dari n buah elemen antena dipol ideal. Jika beda fasa antar elemen α = 0°, dan jarak anter elemen d = λ⁄2, tentukan dan plot pola elemen pola elemen $g(\theta,\phi)$, pola faktor susun $f(\theta,\phi)$ dan pola total $F(\theta,\phi)$ masing-masing untuk jumlah elemen n:
   a) n = 2
      - Pola elemen $g(\theta,\phi)$



      - Pola factor susun $f(\theta,\phi)$

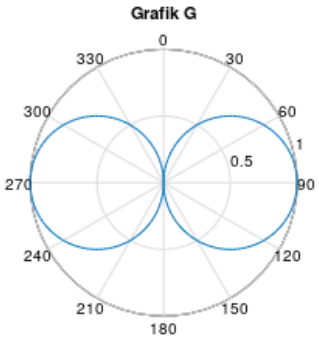


      - Pola total $F(\theta,\phi)$

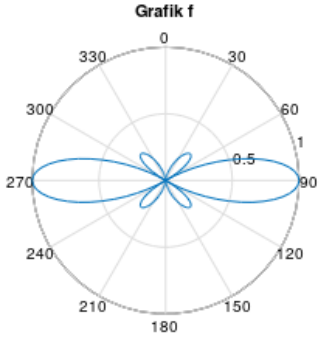


b) n = 4

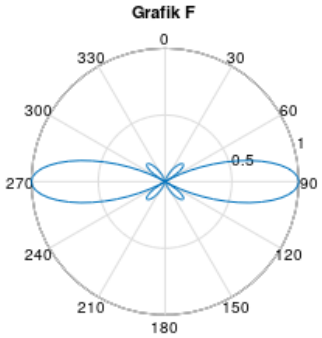
- Pola elemen $g(\theta,\phi)$



- Pola factor susun $f(\theta,\phi)$

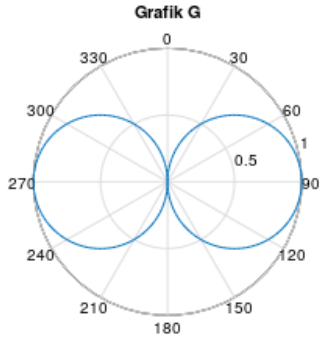


- Pola total $F(\theta,\phi)$

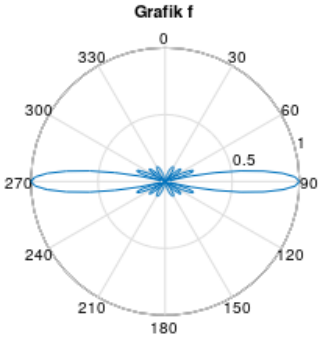


c) n = 8

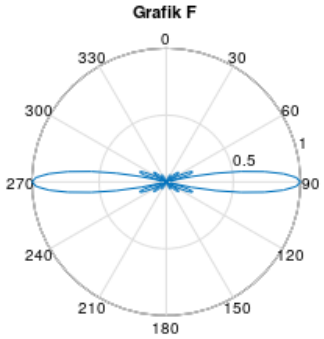
- Pola elemen $g(\theta,\phi)$



- Pola factor susun $f(\theta,\phi)$

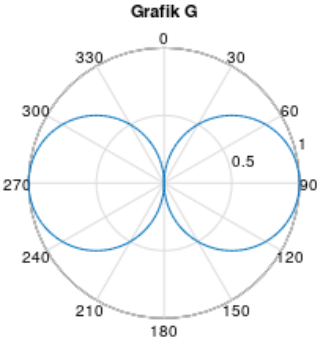


- Pola total $F(\theta,\phi)$

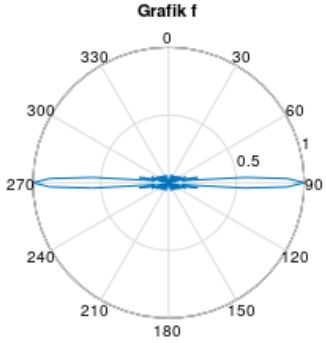


d) n = 16

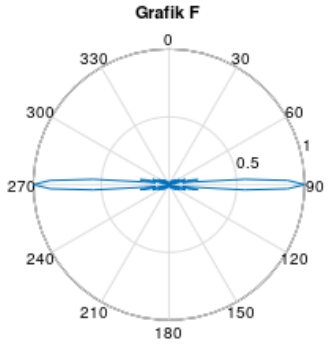
- Pola elemen $g(\theta,\phi)$



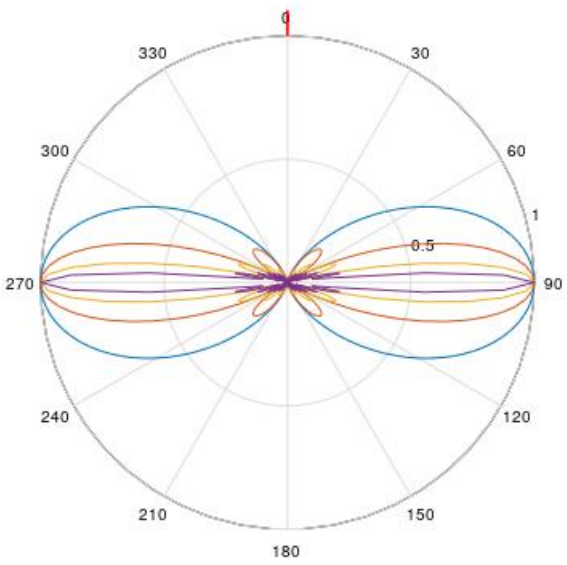
- Pola factor susun $f(\theta,\phi)$



- Pola total $F(\theta,\phi)$



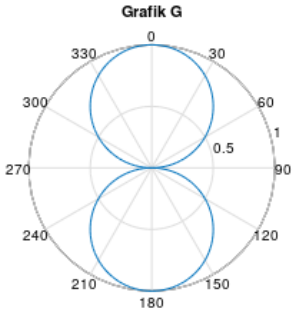
Gambar apabila di plot dalam 1 grafik



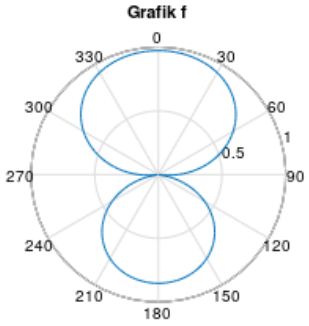
- (Blue) n = 2
- (Red) n = 4
- (Yellow) n = 6
- (Purple) n = 16

4. Sebuah antena susun (array) endfire terdiri dari n buah elemen antena dipol ideal. Antena susun ini diarahkan ke θ = 0°. (Tetapkan pola elemennya adalah $g(\theta,\phi)$ = cos θ). Plot pola faktor susun $f(\theta,\phi)$ dan pola total $F(\theta,\phi)$ masing-masing untuk jumlah elemen n:

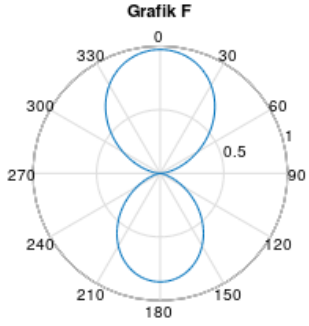
   a) n = 2
      - Pola elemen $g(\theta,\phi)$



      - Pola factor susun $f(\theta,\phi)$

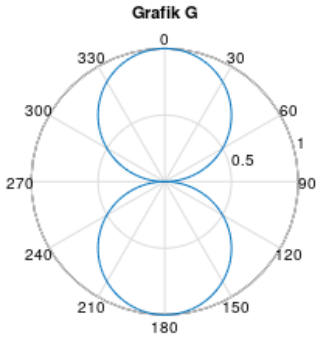


      - Pola total $F(\theta,\phi)$

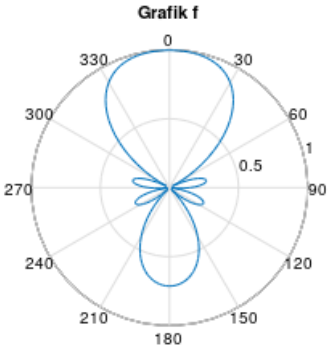


b) n = 4

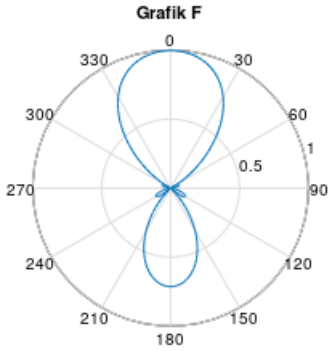
- Pola elemen $g(\theta,\phi)$



- Pola factor susun $f(\theta,\phi)$

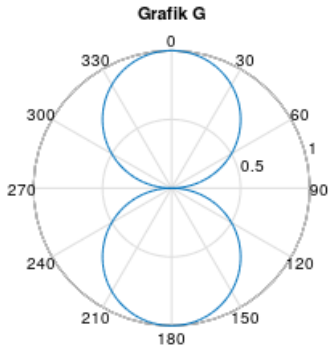


- Pola total $F(\theta,\phi)$

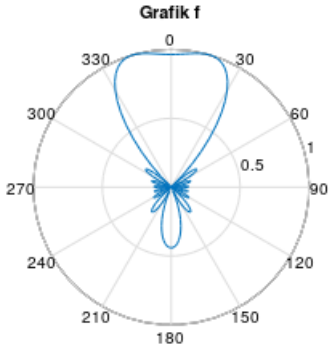


c) n = 8

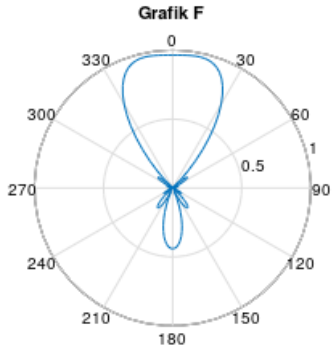
- Pola elemen $g(\theta,\phi)$



- Pola factor susun $f(\theta,\phi)$

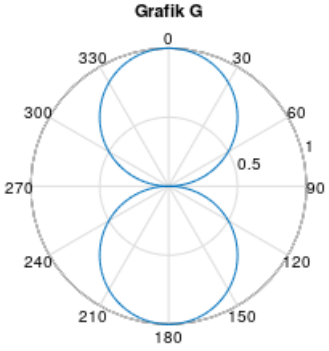


- Pola total $F(\theta,\phi)$

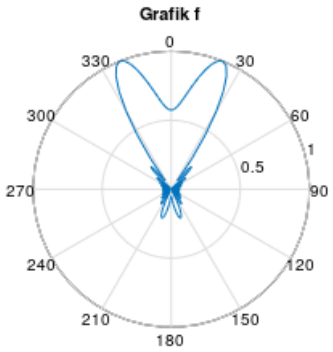


d) n = 16

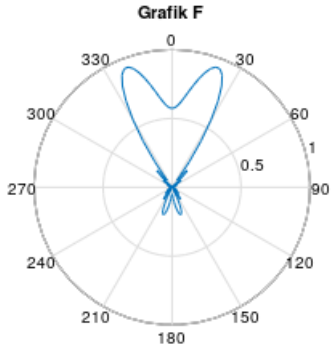
- Pola elemen $g(\theta,\phi)$



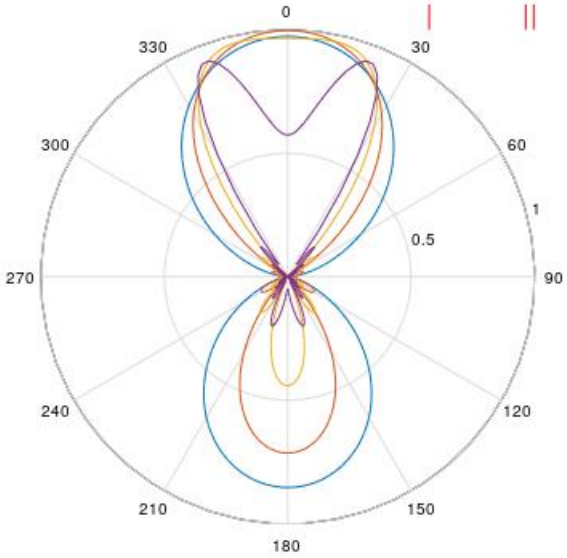
- Pola factor susun $f(\theta,\phi)$



- Pola total $F(\theta,\phi)$



Gambar apabila di plot dalam 1 grafik



- (Blue) n = 2
- (Red) n = 4
- (Yellow) n = 6
- (Purple) n = 16